

PROJECT PROPOSAL
DIGITAL IMAGE PROCESSING
S1 SAINS DATA - UNIVERSITAS NEGERI SURABAYA

PROJECT IDENTITY	
Title	Alzheimer's Disease Severity Classification from Brain MRI Using Hybrid Gabor, LBP, and GLCM Feature Extraction with SVM and Random Forest
Topic	Alzheimer's Image Classification
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1. INTRODUCTION
1.1. Background
One of the most prevalent neurodegenerative diseases, Alzheimer's disease contributes significantly to the cognitive deterioration that older persons experience. Memory loss, cognitive decline, and behavioural abnormalities that worsen with time are the disease's hallmarks [9]. In order to choose the best course of treatment and stop the disease from getting worse, early detection and grading of the condition's severity are essential. However, traditional diagnostic methods, which mostly rely on clinical evaluations and subjective assessments, might be unreliable, particularly in the early stages of the illness. To precisely determine the severity of Alzheimer's disease, a technology-driven method that offers unbiased and reliable analysis is required. The ability of medical imaging, such as Magnetic Resonance Imaging (MRI), to accurately display brain structures has made it a crucial tool for supporting Alzheimer's diagnosis. MRI is a safe diagnostic technique that produces comprehensive anatomical images, especially when contrast is used to make structures and diseases more visible [2]. MRI image analysis makes it possible to identify morphological changes in the brain, such as the shrinkage of the hippocampus and cerebral cortex, which are associated with the progression of the illness. There are now many opportunities to automate medical picture analysis thanks to advancements in computer vision and machine learning. In order to recover relevant texture characteristics from MRI scans, feature extraction techniques are essential. Therefore, improving the efficacy of classification systems requires selecting the right feature extraction method. Various texture feature extraction methods have been developed to improve the accuracy of medical image classification, including Gabor Filter, Local Binary Pattern (LBP), and Gray Level Co-occurrence Matrix (GLCM). Gabor filter is effective in capturing frequency and orientation information of texture, while LBP excels in representing local patterns simply and efficiently. Texture feature extraction methods such as LBP have been proven effective in improving image classification accuracy when combined with Support Vector Machine (SVM) [6]. In addition, the development

of the LBP method has also been widely used in medical image classification, including for brain image analysis [5]. On the other hand, GLCM is able to describe spatial relationships between pixels through statistical features such as contrast and homogeneity, and is widely used in texture-based classification [7]. Although each method has its advantages, the use of a single method is often insufficient to represent the complexity of texture in brain MRI images. Therefore, a hybrid approach that combines the three methods has the potential to produce a more comprehensive feature representation.

The categorisation stage is an important step in assessing the severity of Alzheimer's after feature extraction. Because they can handle complex and high-dimensional data, algorithms like Support Vector Machine (SVM) and Random Forest are commonly employed. While Random Forest offers benefits in interpretability through feature importance analysis, SVM with RBF kernel can separate data non-linearly [8]. Additionally, model generalisation is enhanced by the application of validation techniques like stratified cross-validation and hyperparameter tuning. The robust performance of the final model is further ensured by performance evaluation utilising metrics like accuracy, F1-score, and AUC-ROC. The suggested approach is anticipated to produce precise and trustworthy results in classifying Alzheimer's severity when feature extraction techniques and classification algorithms are properly combined.

In contemporary medical image analysis, combining different feature extraction techniques into a single hybrid framework is becoming a more and more important strategy. The system can more thoroughly capture texture information from both local patterns and global spatial relationships because of the combination of Gabor, LBP, and GLCM features. This method not only enhances the quality of data representation, but it also makes it easier for the model to differentiate between Alzheimer's severity levels. Additionally, maintaining the stability and consistency of system performance depends heavily on feature normalisation and model validation. This research is anticipated to significantly advance the creation of a more precise, impartial, and trustworthy MRI-based categorisation system with an organised and thorough process design.

1.2. Problem-Solving Approach

In order to overcome the limits of traditional diagnosis in identifying Alzheimer's severity, which is often subjective, this research suggests a medical image processing and machine learning-based approach. To ensure a balanced class distribution, the procedure starts with data preparation utilising a stratified split and stratified 5-fold cross-validation. Preprocessing includes scaling the pictures to 128 by 128 pixels, turning them to greyscale, using CLAHE to improve contrast, and applying Gaussian filtering to reduce noise. Additionally, Otsu's thresholding is used in region-of-interest (ROI) segmentation to isolate key regions from MRI images. To add diversity and decrease overfitting, data augmentation techniques are used to the training data. Before moving on to the other feature extraction and modelling phases, our method guarantees the best possible data quality.

A hybrid method that combines Gabor Filter, Local Binary Patterns (LBP), and Gray-Level Co-occurrence Matrix (GLCM) is used to extract features. LBP is used for local patterns, GLCM is used for spatial interactions between pixels, and Gabor is used to record frequency and orientation information. The resulting features are then

combined and normalized using StandardScaler to achieve a balanced scale. The classification process uses a Support Vector Machine (SVM) with an RBF kernel as the main model and Random Forest as a comparison. Evaluation is carried out using accuracy, precision, recall, weighted F1-score, AUC-ROC, and confusion matrix. To improve interpretability, SHAP, feature importance, and PCA or t-SNE visualization are used. This approach is expected to produce an accurate, robust, and reliable classification system and can contribute to the development of medical image-based diagnostic systems.

2. METHODOLOGY

2.1. Dataset Exploration

The dataset used in this research is the Alzheimer's Disease Multiclass Images Dataset [1], publicly available on Kaggle. This dataset contains 44.000 skull-stripped brain MRI images categorized into four distinct classes based on Alzheimer's disease severity: Non Demented, Very Mild Demented, Mild Demented, and Moderate Demented. This dataset is an unsampled version of a prior heavily imbalanced dataset, designed to support balanced multi-class classification.

Table 1. Dataset Overview

Attribute	Detail
Source	Kaggle [1]
Format	JPG
Total Images	+ - 44.000
Number of Classes	4
Image Type	Brain MRI, skull-stripped

The detailed distribution of images across each class is described in Table 2 below.

Table 2. Class Description

Class Label	Abbreviation	Number of Images	Description
Non Demented	ND	12.800	No signs of dementia, serves as the healthy baseline
Very Mild Demented	VMD	11.200	Early signs with very mild symptoms
Mild Demented	MD	10.000	Clear signs but still at mild stage
Moderate Demented	MOD	10.000	More pronounced moderate-severity symptoms

The dataset is relatively balanced across all four classes, ranging from 10.000 to 12.800 images per class. This balanced distribution is the result of intentional upsampling from the original heavily imbalanced version, and it supports the use of

stratified splitting (70/15/15) and stratified 5-fold cross-validation in this research to maintain proportional class representation across all data subsets. All skull-stripped images align directly with the grayscale conversion and ROI cropping steps in the preprocessing pipeline.

2.2. Research Step

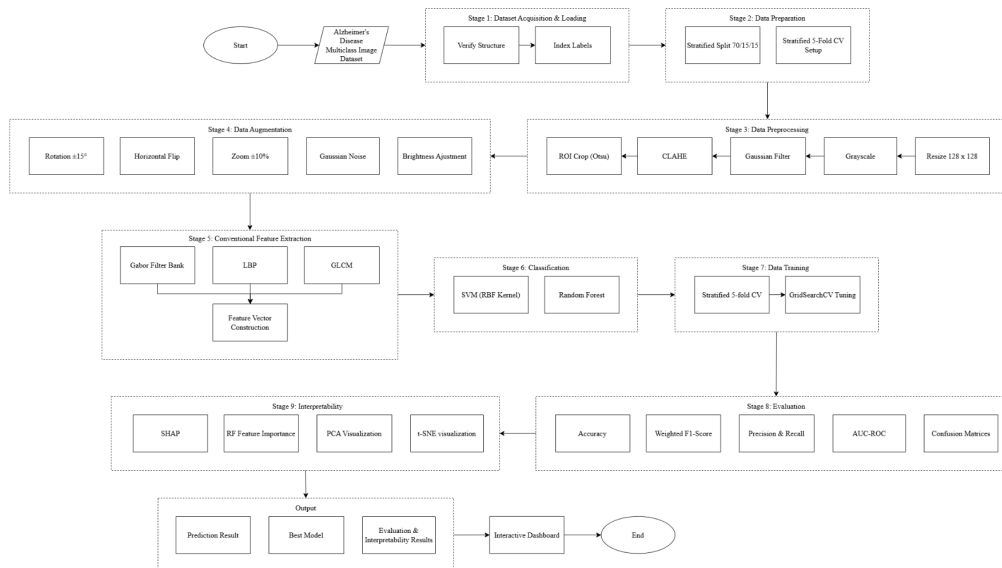


Figure 1. Research Flowchart

According to Figure 1, the research is executed across nine stages to ensure methodological coherence, reproducibility, and clinical relevance in classifying Alzheimer's disease severity from brain MRI images.

1. Stage 1: Dataset Acquisition and Loading

- Download dataset [1] from Kaggle and load into the environment.
- Verify directory structure for all 4 class folders: Non Demented, Moderate Demented, Mild Demented, and Very Mild Demented.
- Remove corrupted images; index-label image paths before splitting.

2. Stage 2: Data Preparation

- Stratified splitting (70% train, 15% validation, and 15% test).
- Stratified 5-fold cross-validation to improve reliability and reduce evaluation variance.

3. Stage 3: Data Preprocessing

Each raw MRI image is preprocessed sequentially.

Table 3. Preprocessing Steps and Purposes

Step	Method	Purpose
1.	Resize to 128×128 px	Standardize spatial dimensions for consistent feature computation
2.	Grayscale Conversion	Reduce unnecessary color complexity, MRI diagnostic info is captured in intensity values
3.	Gaussian Filtering	Suppress high-frequency noise while preserving tissue boundaries

4.	Contrast Limited Adaptive Histogram Equalization (CLAHE)	Enhance local contrast in low-contrast regions, and make subtle cortical texture differences more distinguishable, especially between the Mild and Very Mild classes [10]
5.	Region of Interest (ROI) cropping via Otsu's Thresholding	Isolate brain tissue from black background; focus extraction on diagnostically relevant regions

4. Stage 4: Data Augmentation

Applied to training data only to improve generalization:

- Random rotation ($\pm 15^\circ$)
- Horizontal flipping
- Zoom ($\pm 10\%$)
- Gaussian noise injection
- Brightness adjustment

5. Stage 5: Conventional Feature Extraction

3 conventional feature extraction methods are applied in parallel to each preprocessed image, then combined into a single normalized feature vector.

Table 4. Feature Extractions' Configuration and Output

Method	Configuration	Output
Gabor Filter Bank	8 orientations (0° - 157.5°), 5 frequency scales per image	4 descriptors per response: mean, variance, energy, entropy
Local Binary Pattern (LBP)	Multi-radius: $r = 1, 2, 3$ with 8 sampling points per radius	Normalized uniform LBP histograms per-radius, concatenated
Gray-Level Co-occurrence Matrix (GLCM)	Co-occurrences matrices at 4 angles (0° , 45° , 90° , and 135°)	4 Haralick features per matrix: contrast, correlation, energy, homogeneity
Feature Vector Construction	Concatenation of all Gabor + LBP + GLCM features	~300-dim normalized vector (StandardScaler)

- Gabor is selected because it captures orientation-sensitive edge and ridge patterns in cortisol tissue that progressively degrade with Alzheimer's severity [3].
- LBP is selected because it encodes local microstructural tissue patterns at multiple spatial scales and is robust to illumination variation and minor rotational differences that are commonly found between MRI scans of different subjects [4].

- GLCM is selected because it captures the spatial intensity relationship between neighboring pixel intensities and reflects cortical structural coherence, which diminishes with neurodegeneration [10].
- Feature Vector Construction then constructed and resulting vector of ~ 300 dimensions, then being normalized using StandardScaler to eliminate scale dominance from any single feature subspace.

6. Stage 6: Classification

The normalized feature vector is independently fed into 2 classifiers.

Table 5. Role Comparison of SVM with RBF Kernel and Random Forest

Model	Role	Configuration
SVM with RBF Kernel	Primary classifier	Hyperparameters C and γ tuned via GridSearchCV
Random Forest	Comparator classifier	n_estimators and max_depth tuned via GridSearchCV

- SVM (RBF Kernel) as the primary model because it is effective in high-dimensional and non-linear feature spaces.
- Random Forest as the comparator model because it enables feature importance analysis in the interpretability stage.

7. Stage 7: Data Training

The classification results are then being trained.

- Both classifiers are trained under stratified 5-fold cross-validation to ensure each fold contains proportional class representation.
- GridSearchCV is used for systematic hyperparameter optimization across both models, minimizing the risk of suboptimal performance because of default parameter settings.

8. Stage 8: Evaluation

Model performance is evaluated using the following metrics.

Table 6. Evaluation Metrics and Their Purposes

Metric	Purpose
Accuracy	Overall classification correctness
Weighted F1-Score	Class-level precision-recall balance
Precision and Recall	Per-class analysis, important for early-stage classes
Area Under the Receiver Operating Characteristic Curve (AUC-ROC)	Per-class discriminative capability across decision thresholds
Confusion Matrix	Inter-class misclassification patterns between neighboring severity levels

9. Stage 9: Interpretability

3 methods applied to ensure accurate, transparent, and clinically meaningful results.

Table 7. Methods Used in Interpretability

Method	Scope	Output
SHapley Additive exPlanations (SHAP)	Global + local	Feature contribution per prediction to identify most influential features per class
RF Feature Importance	Global	Ranks all features by aggregate discriminative contribution across the full dataset
Principal Component Analysis (PCA) Visualization	Global	Projects ~300 dimensions feature space into 2D to validate class separability
t-distributed Stochastic Neighbor Embedding (t-SNE) Visualization	Global	Projects feature space into 2D by highlighting cluster cohesion

After generating the prediction, best model, evaluation, and interpretation results are generated, an interactive dashboard is constructed to classify Alzheimer's severity.

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